

AI-assisted pharmacometric analysis workflow

R/Pharma 2026

Mutaz Jaber

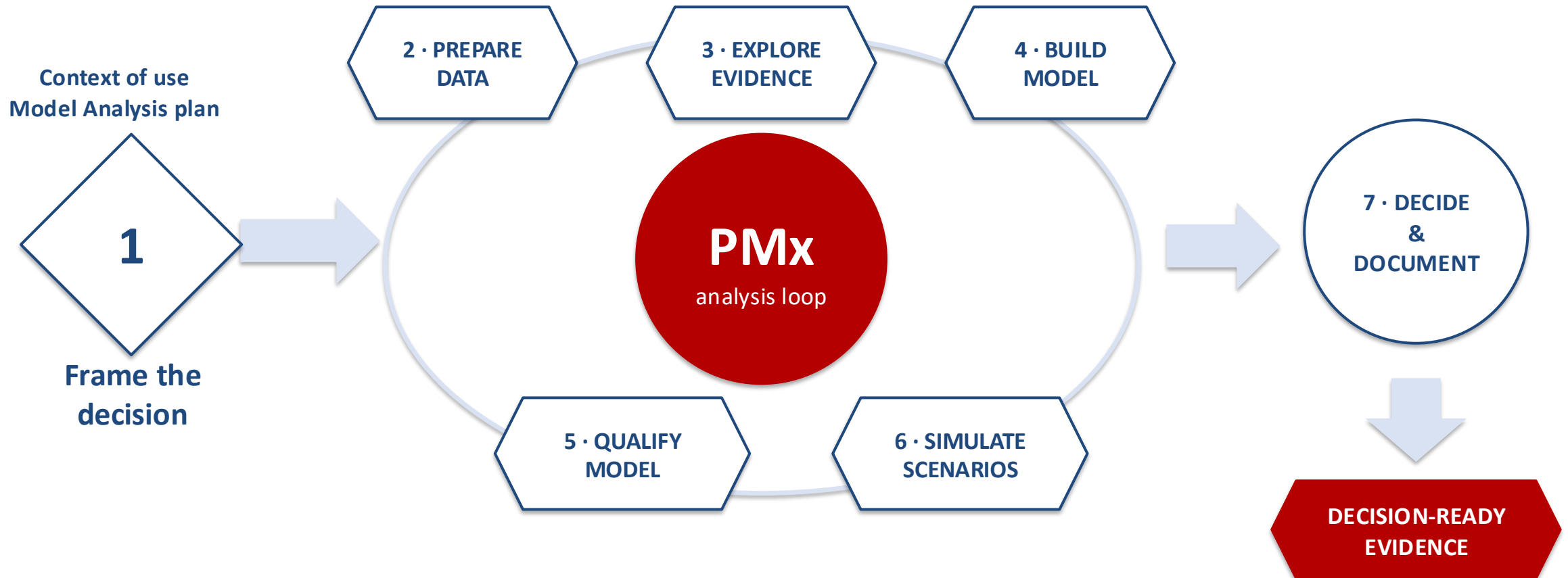
On behalf of the **nlmixr** development team:

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General workflow of a typical pharmacometric analysis

An iterative workflow: evidence, assumptions, and decisions are revisited as data mature.



Sources: ICH M15 (2026); FDA Population PK Guidance (2022); EMA PopPK Reporting Guideline (2007).

How AI fits in this workflow?

Three properties of pharmacometric work make it well suited to AI drafting — under expert review.

STANDARDIZED FLOW

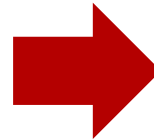
Every analysis moves through the same defined stages and outputs.

REPETITION

The same steps recur across studies, compounds and programs.

REVIEW CULTURE

Nothing ships without expert QC and documented sign-off.



This is why AI is good at providing draft analysis

A pattern to learn. Predictable structure the model can follow.

Volume to absorb. Repeated setup becomes an automated first pass.

A safety net. Experts review every draft before it counts.

How Agent SKILLS bring benefits to the flow

A reusable, checked-in instruction set that teaches the agent your conventions

REUSABLE

Written once, invoked on every study, compound and program.

Version-controlled

Versioned in the repository; reviewed and updated like code.

Guidelines encoded

Units, data coding, naming, output style and QC rules.

THE PHARMACOMETRIC WAY

Our accepted method for the task, not the model's improvisation.

WITHOUT A SKILL

Prompted from scratch each time — plausible output that must be re-checked line by line.

WITH A SKILL

Same procedure, same structure, same QC gates — output a reviewer can trust and trace.

SAME TASK • SAME CONVENTIONS • EVERY STUDY • ANY TEAM MEMBER

SKILL.md is the important file to start with

```
.posit/assistant/skills/pk-nca/  
├── SKILL.md           # Required: metadata + instructions  
├── scripts/          # Optional: executable code  
├── references/        # Optional: documentation  
├── assets/           # Optional: templates, resources  
└── ...               # Any additional files or directories
```

SKILL.md

name: **pk-nca**

description: Runs a non-compartmental analysis (NCA) of PK data in R with the PKNCA package. Use when the user asks for NCA, PKNCA, or PK parameters from concentration-time data.

Running Non-compartmental analysis

....

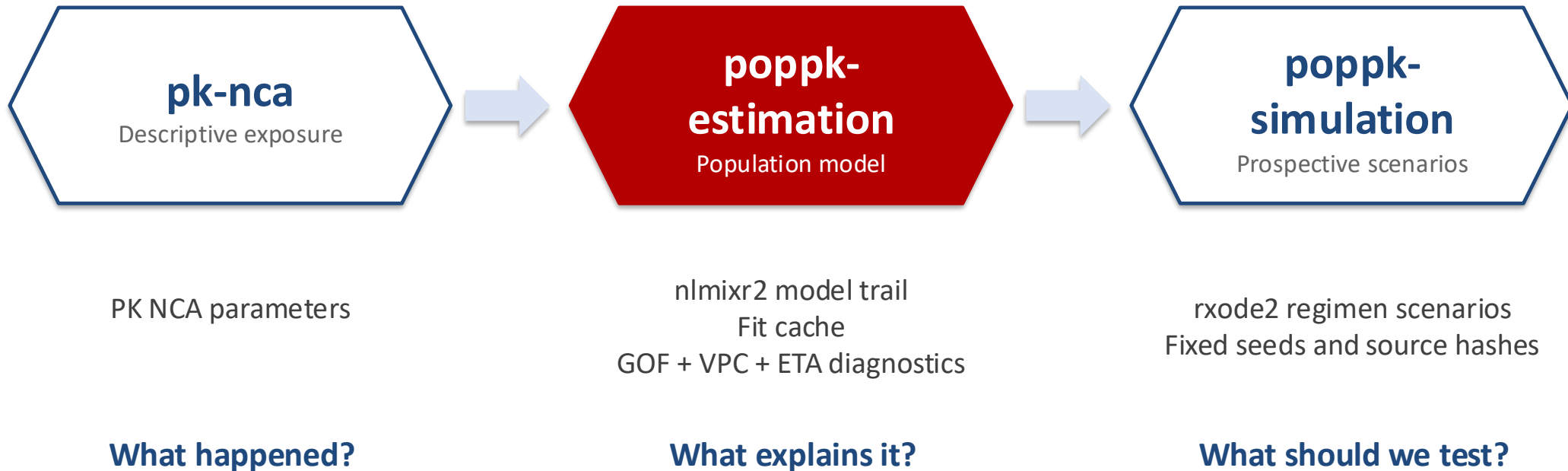
LLMs add value when they orchestrate validated pharmacometrics skills

Natural-language intent becomes reproducible analysis through governed domain tools and expert review.



Proposed pharmacometric skills

Each skill owns one computational step, then packages traceable tables, figures, and reporting for review.



SHARED OUTPUT SYSTEM

TABLES • FIGURES • REPORTING | VERSIONED CODE + DATA • LOGS • HASHED PROVENANCE • QC GATES

Detailed live demos follow in this order

Example prompt: launch a complete pk-nca analysis

pk-nca

Suggested prompt:

Set up study XYZ-222: "Phase 1 SAD of XYZ-222", concentrations in ng/mL, time in h, dose in mg. Then run NCA for data/xyz222-pk.csv (oral) and tell me whether it is ready for QC.

WATCH THE WORKFLOW

1 Inspect

Review the dataset before mapping EVID/CMT, times, dose and units.

2 Version

Fill project.yaml and scaffold a new NCA version.

3 Execute

Run PKNCA once; build the database, tables, figures and specification.

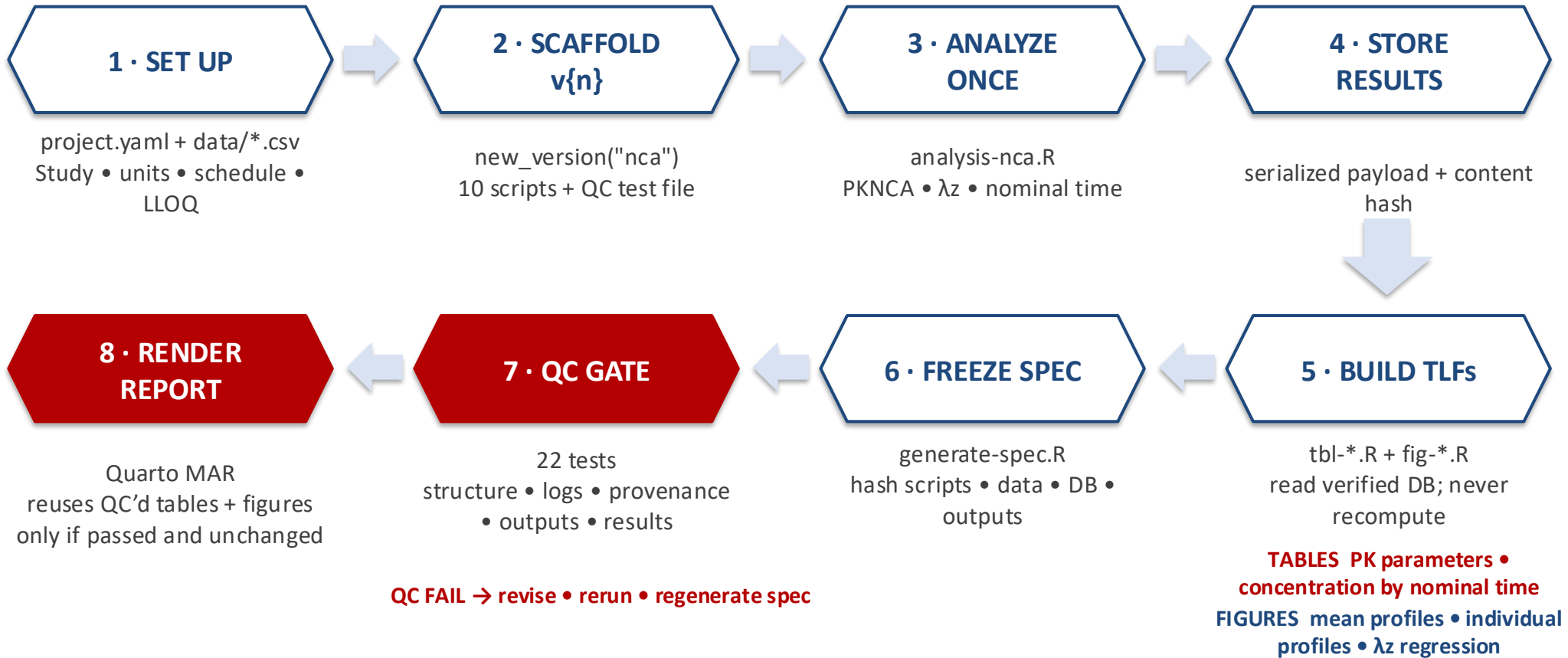
4 Hand off

Show key parameters, flagged λ_z fits and the 22-test QC result.

EXPECTED ENDPOINT A versioned, hash-verified NCA package that is ready—or clearly not ready—for reviewer QC.

Live demo: pk-nca creates a hash-verified, QC-gated NCA report

Prompt Execution



COMPUTE ONCE • READ EVERYWHERE • LOG EVERY RUN • ONE NOMINAL-TIME DEFINITION • CHANGE = NEW VERSION

Example prompt: develop and qualify a first popPK model

poppk- estimation

Suggested prompt:

Develop a popPK model for data/xyz222-pk.csv. Start with a one-compartment model with first-order absorption and proportional error model. Use NCA results as initial estimates

WATCH THE WORKFLOW

1 Inspect

Confirm EVID/CMT coding, dosing history, covariates and analysis units.

2 Version

Scaffold a popPK version and define run001 plus its rationale.

3 Execute

Fit with nlmixr2; cache the run; build diagnostics, tables and the model trail.

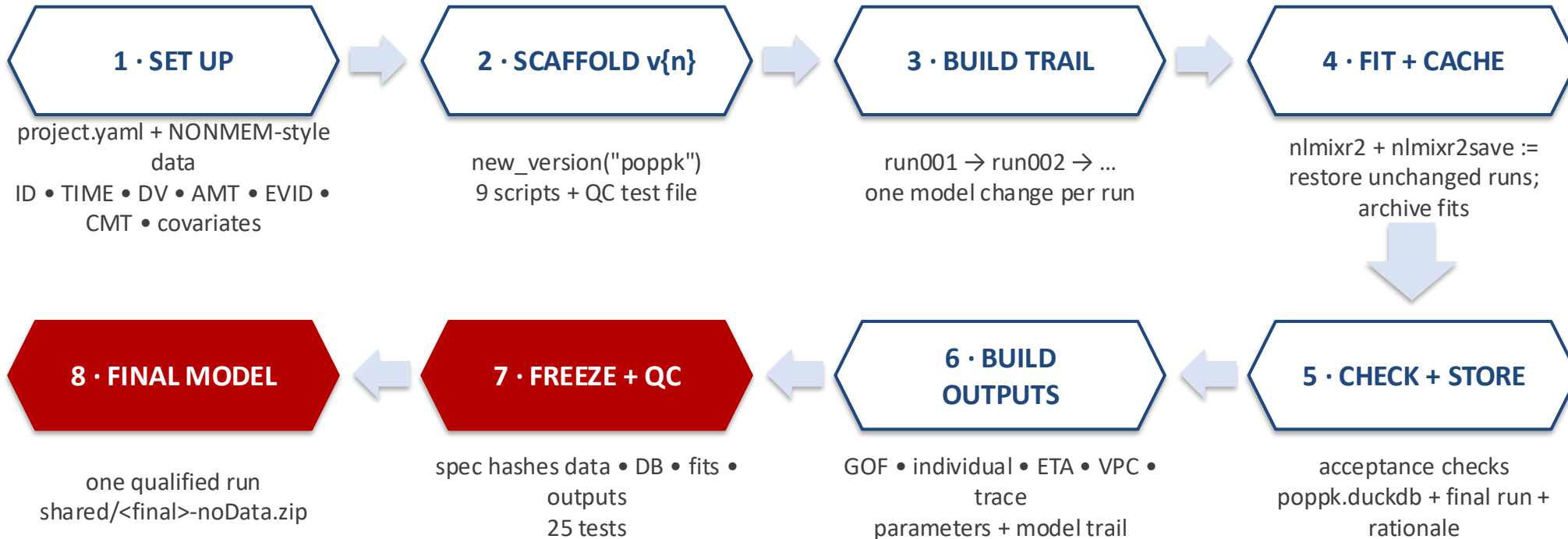
4 Hand off

Report estimates, acceptance warnings, final rationale and the 25-test QC result.

EXPECTED ENDPOINT A qualified final run with a transparent model-building trail and a shareable no-data fit archive.

Live demo: popPK builds a cached model trail into a QC-ready final model

Prompt Execution



BLOCKING CHECKS finite OFV • covariance • structural SE • bounds • residual error

WARNINGS structural %RSE > 50 • ETA shrinkage > 30% • BSV outside 1–100%

COMPUTE ONCE • SAME METHOD ACROSS TRAIL • FIXED SEEDS • LOG EVERY RUN • CHANGE = NEW VERSION

Example prompt: compare regimens with reproducible popPK simulations

poppk-
simulation

Suggested prompt:

Using the popPK v1 final model, simulate 320 mg q12h, 480 mg q12h and 640 mg q24h for 7 days, with 100 subjects × 10 studies. Compare steady-state C_{max}, C_{min} and AUC_{tau}.

WATCH THE WORKFLOW

1 Source

Load the final popPK model and record its database and specification hashes.

2 Scenarios

Define three rxode2 regimens, steady-state windows and a fixed seed.

3 Execute

Simulate once; derive exposure metrics; build profile and comparison outputs.

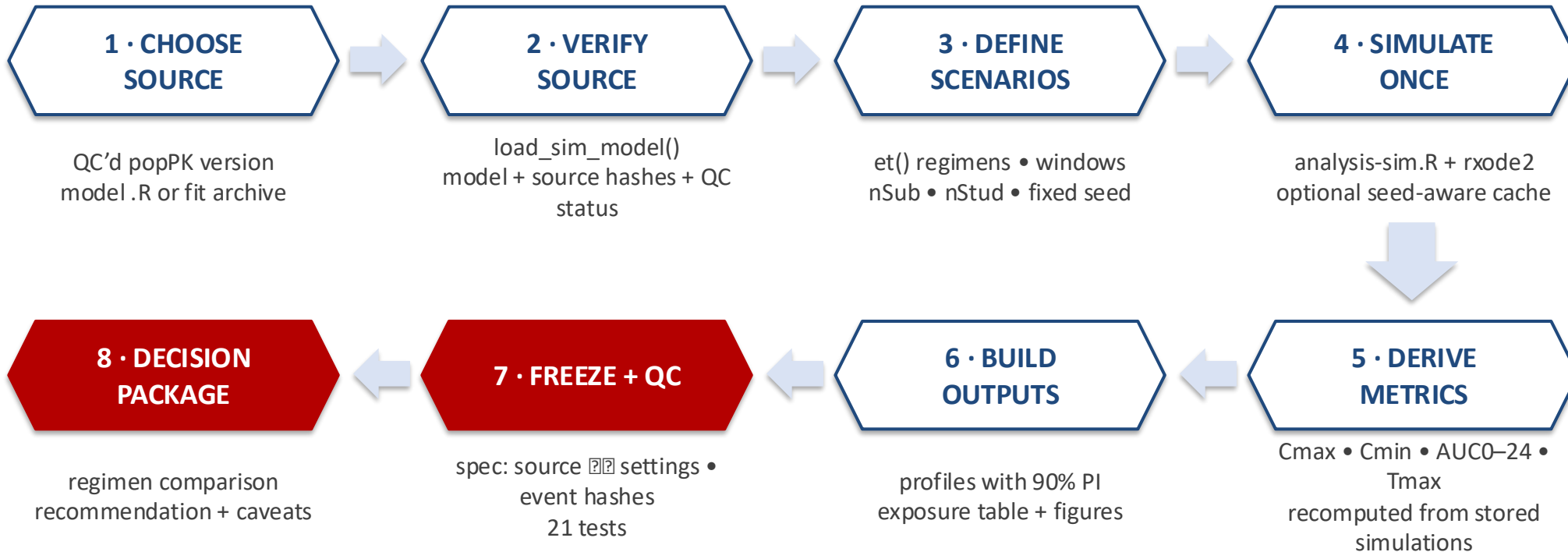
4 Hand off

Show regimen comparisons, caveats and the reproducible 21-test QC result.

EXPECTED ENDPOINT A decision package linked to its model source, scenarios and seed—and invalidated if that source changes.

Live demo: popPK simulations remain linked to their model source and seed

Prompt: “Simulate candidate regimens from the final model, summarize steady-state exposure, and verify that the decision package is reproducible.”



QC RE-DERIVES METRICS AND RE-SIMULATES THE FIRST SCENARIO

SOURCE REFIT OR SPEC CHANGE → SIMULATION QC FAILS UNTIL RE-RUN

SOURCE HASH LINK • FIXED SEEDS • REPRODUCIBLE FIRST SCENARIO • CHANGE IN SOURCE = RERUN SIMULATION

File
Edit
Code
View
Plots
Session
Build
Debug
Profile
Tools
Help

Go to file/function
Addins

Posit Assistant
R 4.6.1

Posit Assistant

Start a conversation with Posit Assistant to analyze data, create visualizations, write code, and more.

Posit Assistant is a data-science centric AI agent. It has the capability to read, analyze, and modify your data, as well as execute arbitrary code on your behalf. Always verify results, as AI assistants can sometimes produce incorrect code.

Ask Posit Assistant...
OK · \$0.00
GLM 5.3 Flash (Low)

Untitled1
1

Run
Source

1:1 (Top Level)
Posit AI: No completions available.
R Script

Console
Terminal
Background Jobs

R 4.6.1 · /cloud/project/
>

Environment
History
Connections
Tutorial

Files
Plots
Packages
Help
Viewer
Presentation

New Folder
New File
Upload
Delete
Rename
More

Cloud > project

Name	Size	Modified
..		
.posit		
.Rhistory	0 B	Sep 29, 2026 at 7:30 PM
AGENTS.md	9 KB	Sep 29, 2026 at 1:59 PM
CLAUDE.md	11 B	Sep 29, 2026 at 1:59 PM
data		
docs		
examples		
materials		
model		
project.Rproj	205 B	Sep 30, 2026 at 8:22 AM
project.yaml	1.7 KB	Sep 29, 2026 at 1:59 PM
README.md	4.2 KB	Sep 30, 2026 at 7:35 AM